

60

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What is Six Sigma?

- As per Jack Welch “Six Sigma is a quality program that, when all is said and done, improves your customer’s experience, lowers your costs, and builds better leaders.”
- Six Sigma is simply a method of efficiently solving a problem. Using Six Sigma reduces the amount of defective products manufactured or services provided, resulting in increased revenue and greater customer satisfaction.

What is Six Sigma?

- Six Sigma is a method that provides organizations tools to improve the capability of their business processes. This increase in performance and decrease in process variation lead to defect reduction and improvement in profits, employee morale, and quality of products or services. Six Sigma quality is a term generally used to indicate a process is well controlled (within process limits $\pm 3s$ from the center line in a control chart, and requirements/tolerance limits $\pm 6s$ from the center line).

What exactly does “Six Sigma” mean?

- Six Sigma is named after a statistical concept where a process only produces 3.4 defects per million opportunities (DPMO). Six Sigma can therefore be also thought of as a goal, where processes not only encounter less defects, but do so consistently (low variability). Basically, Six Sigma reduces variation, so products or services can be delivered as expected reliably.

About the term 6σ

- Standard Deviation
 - Degree of dispersion from mean value

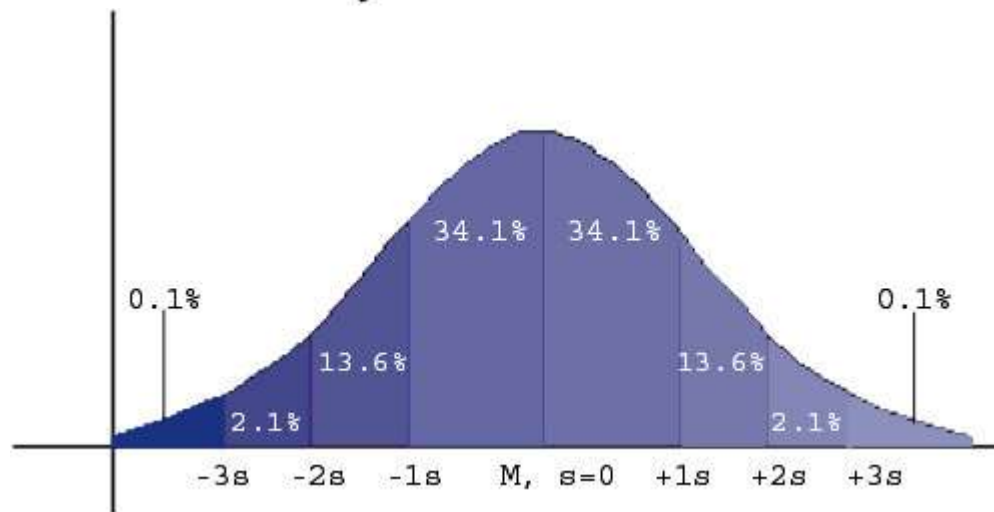
$$s = \sqrt{\frac{\sum (X - M)^2}{n - 1}}$$

s = standard deviation (aka σ)

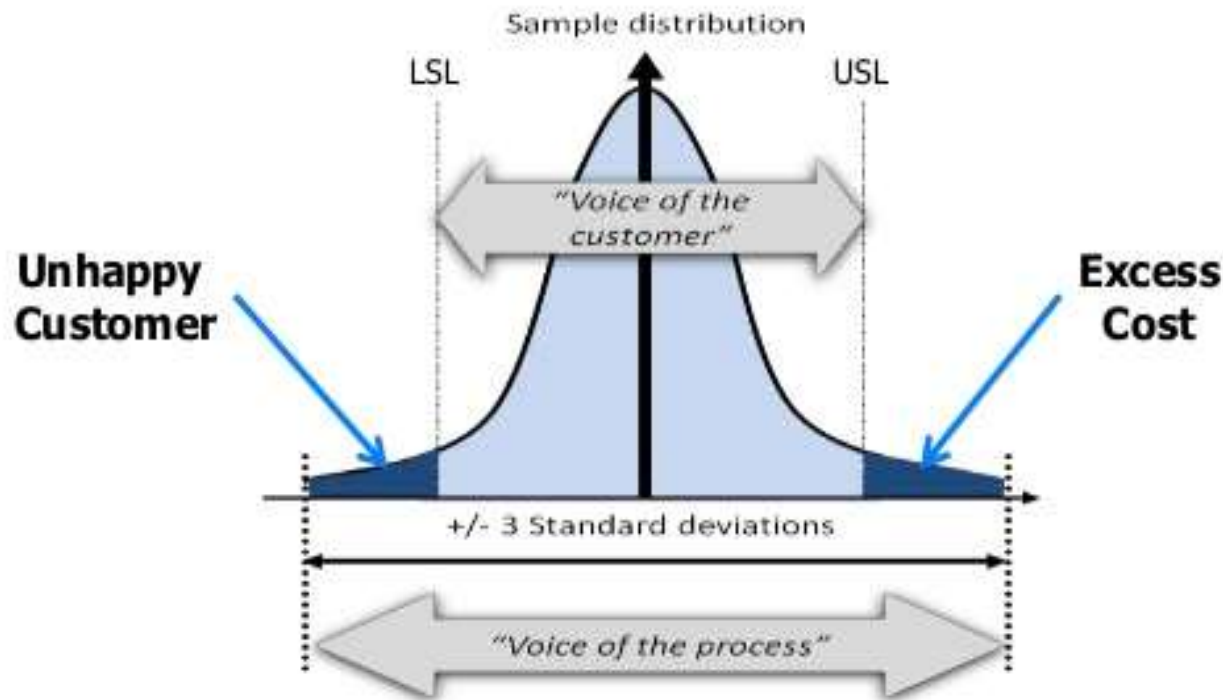
X = data point

M = average of all data points

n = population



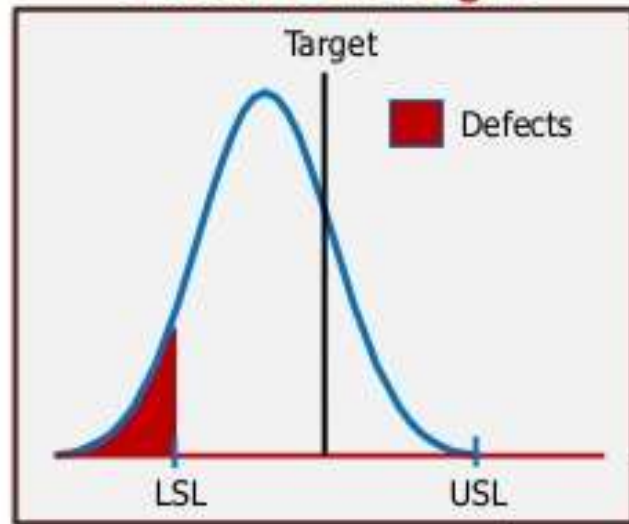
Variability affects your costs and your customers' experience



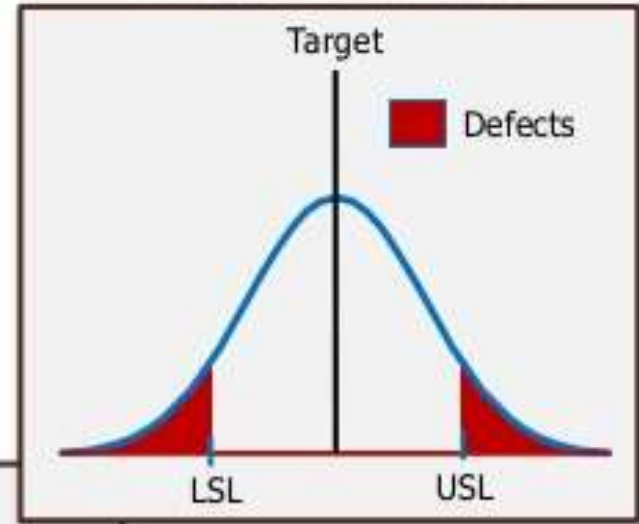
Understanding the nature of process variation is critical since customers feel the effects of process variation as well as the process average

How to Manage Process Variation

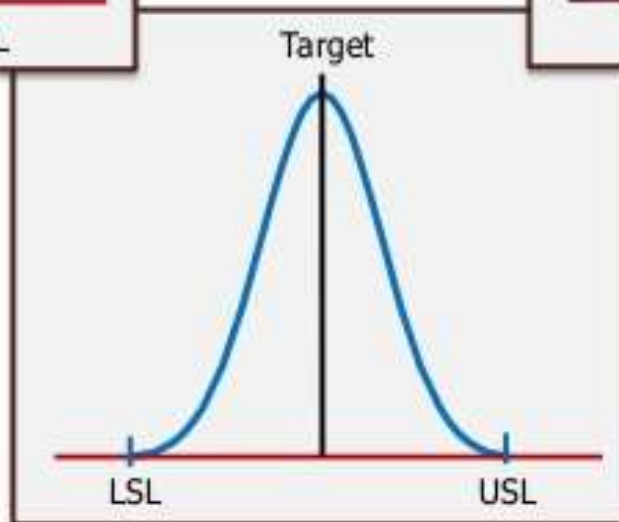
Process Off Target



Excessive Variation



Center Process

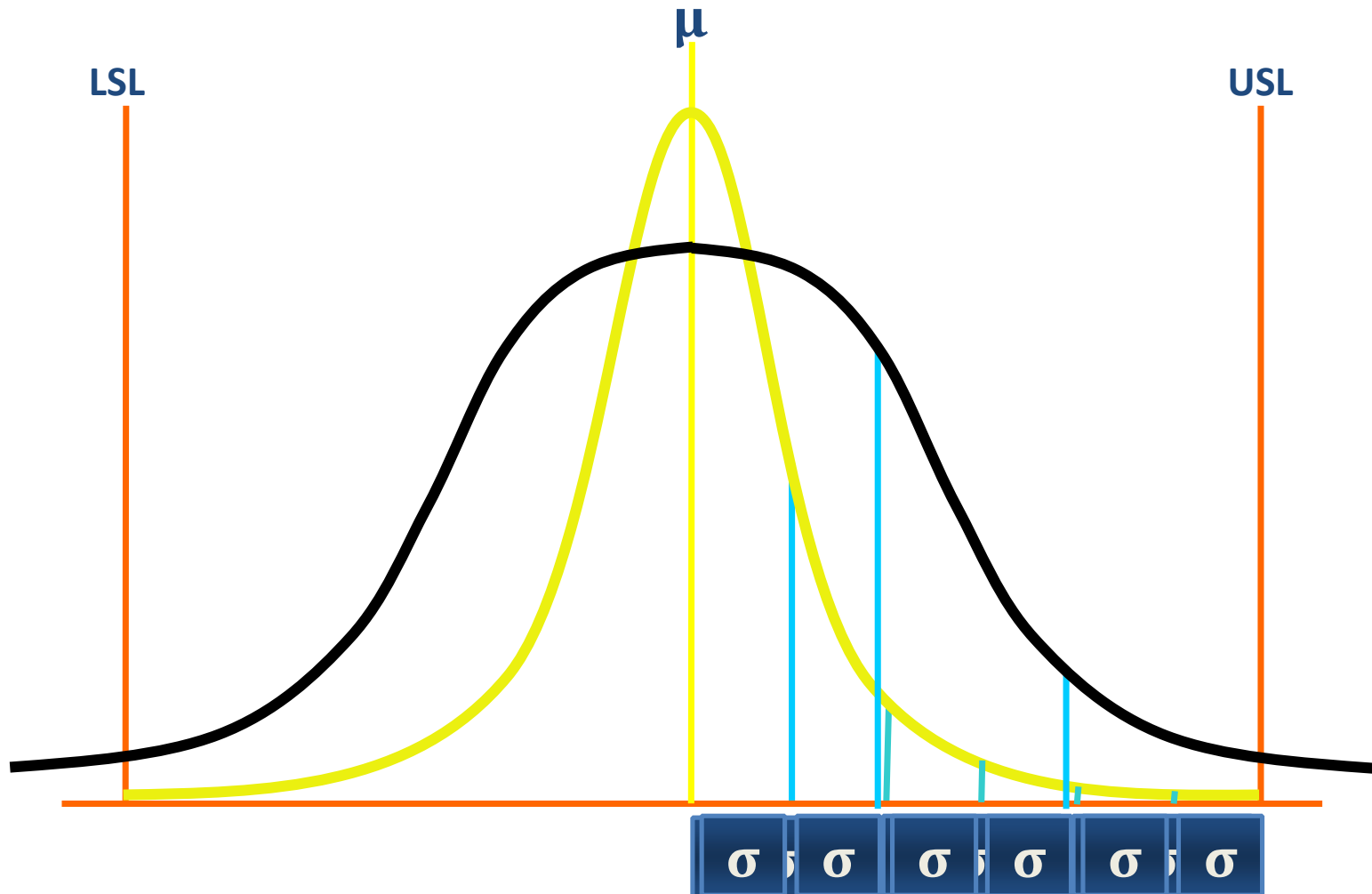


Reduce Spread

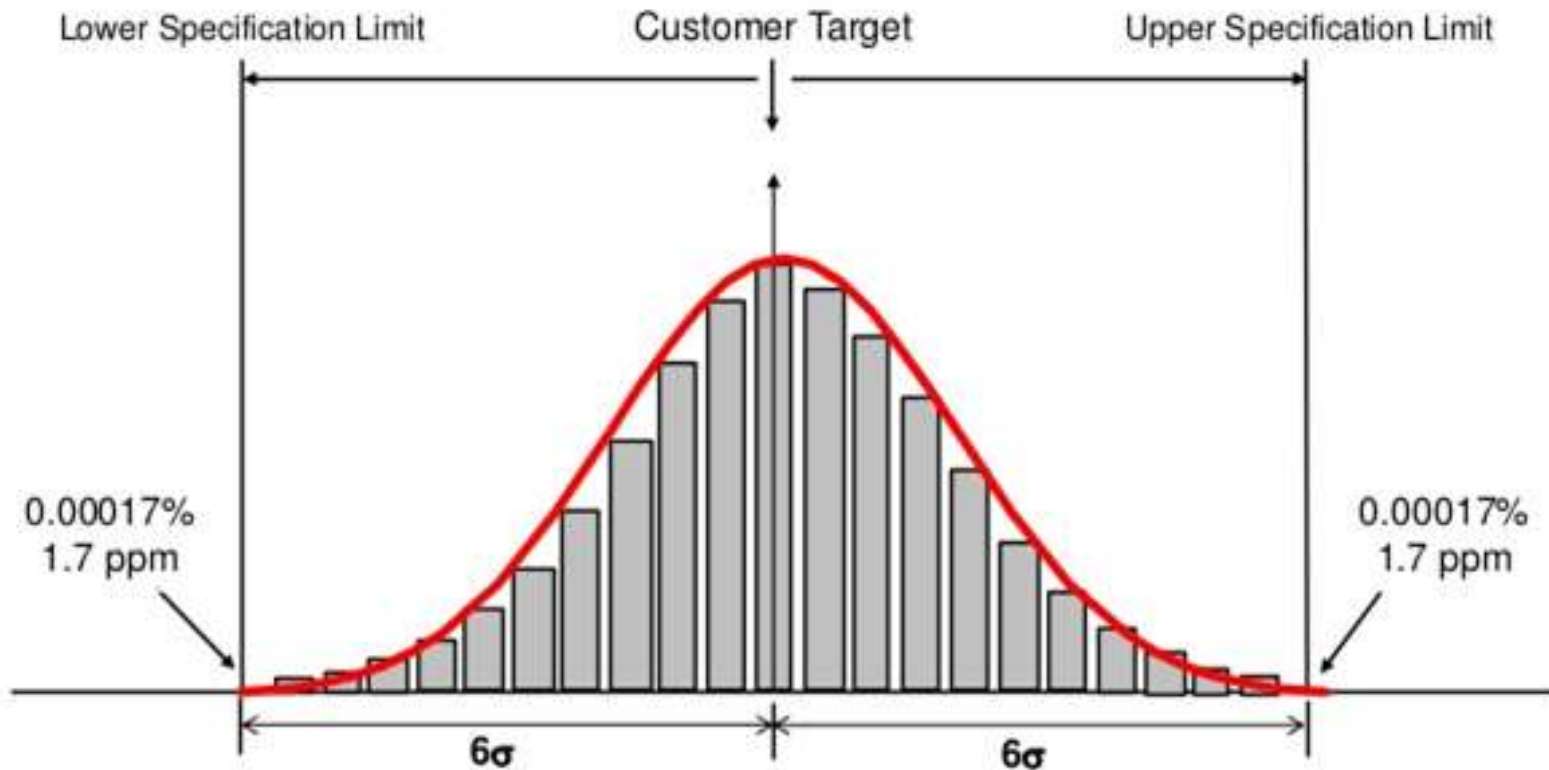


What is “*Six Sigma*”?

99.99966% of values are within specifications



A 6 Sigma Process



= 99.99966% of data inside the limits ($C_p = 2$)

0.00034% of points will be outside of the specification limits, i.e. defects
(= 3.4 parts per million out of spec.)

Real World: What is Sigma Performance?

Is 99% (3.8σ) good enough?

- ➔ 20,000 lost mails per hour
- ➔ Unsafe drinking water almost 15 minutes each day
- ➔ 5,000 incorrect surgical operations per week
- ➔ 2 short or long landings at most major airports daily
- ➔ 200,000 wrong drug prescriptions each year

99.99967% Good – At 6σ

- ➔ 7 lost mails per hour
- ➔ One minute of unsafe drinking water every seven months
- ➔ 1.7 incorrect surgical operations per week
- ➔ One short or long landing at major airports every five years
- ➔ 68 wrong drug prescriptions each year

Features of Six Sigma

- Six Sigma's aim is to eliminate waste and inefficiency, thereby increasing customer satisfaction by delivering what the customer is expecting. Six Sigma follows a structured methodology, and has defined roles for the participants. Six Sigma is a data driven methodology, and requires accurate data collection for the processes being analyzed. Six Sigma is about putting results on Financial Statements. Six Sigma is a business-driven, multi-dimensional structured approach for:
 - Improving Processes
 - Lowering Defects
 - Reducing process variability
 - Reducing costs
 - Increasing customer satisfaction
 - Increased profits

Key Concepts of Six Sigma

- At its core, Six Sigma revolves around a few key concepts.
- **Critical to Quality:** Attributes most important to the customer.
- **Defect:** Failing to deliver what the customer wants.
- **Process Capability:** What your process can deliver.
- **Variation:** What the customer sees and feels.
- **Stable Operations:** Ensuring consistent, predictable processes to improve what the customer sees and feels.
- **Design for Six Sigma:** Designing to meet customer needs and process capability.

Our Customers Feel the Variance, Not the Mean. So Six Sigma focuses first on reducing process variation and then on improving the process capability.

Tools for Six-Sigma and Quality Improvement

- **Elementary and advanced statistics**
- **Product design and reliability analysis**
- **Measurement**
- **Process control & Process improvement**
- **Implementation and teamwork**
- **Customer survey and feedback**
- **Lean thinking**

Examples of 6σ

- Geico: 97% customer satisfaction
~ 4σ
- USPS: 95% 1st class mail delivered on time
~ 3σ

6σ

Can be applied to any industry, service, or
approval rating

Six Sigma Methodologies

- Two key methodologies
 - DMAIC
 - Used for improving existing processes
 - DMADV
 - Used for creating new product/process designs
 - Used for already optimized processes (with DMAIC or another method) that still fall short of expectations

Six-Sigma Problem Solving Approach

1. Define
2. Measure
3. Analyze
4. Improve
5. Control

The logo for the DMAIC (Define, Measure, Analyze, Improve, Control) process. It consists of the letters 'DMAIC' in a bold, sans-serif font. The letters are filled with a vertical gradient, transitioning from a bright yellow at the top to a deep orange at the bottom.

DMAIC

D
M
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C

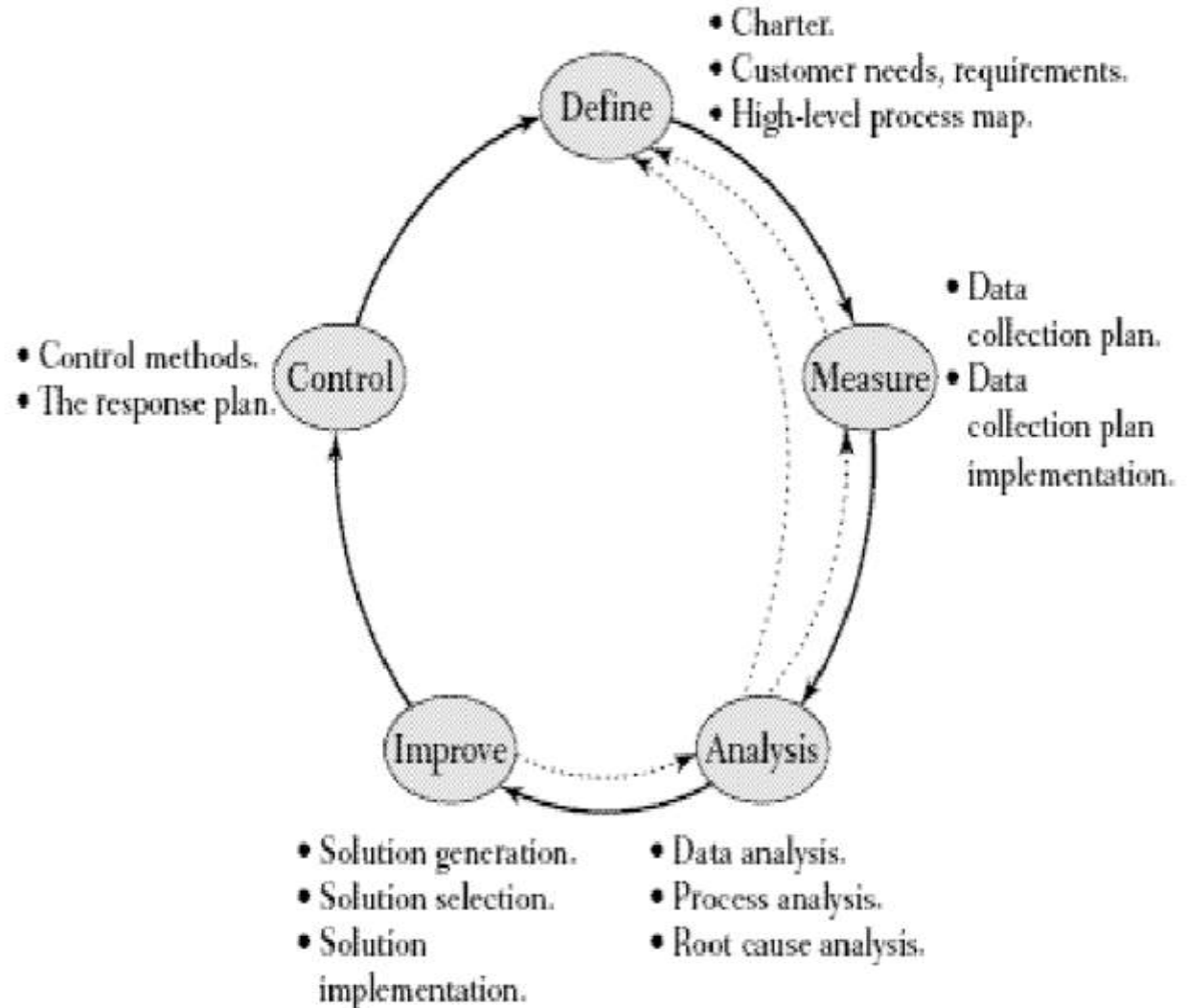
efine

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ontrol



DMAIC: Define

- Define process improvement goals
 - Why the 6 σ program is in place?
- Define customers needs
 - Need vs. requirements to fulfill need
- Create high level process map

DMAIC: Measure

- Measure current process and collect relevant data
- Develop data collection plan
- Collect data from many sources to determine types of defects
- Compare to customer surveys
- Determine shortfalls

DMAIC: Measure

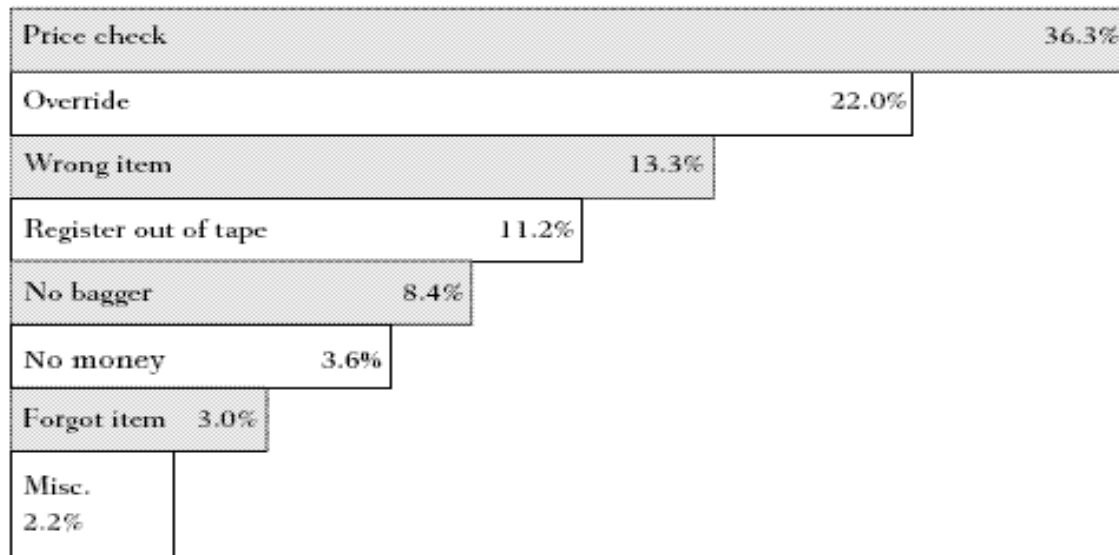
- Determine unit, defect, opportunity
 - Unit = Value of process, input, or output
 - Defect = Something wrong with a unit
 - Too large
 - Too small
 - Not equal to
 - Opportunity = Way to fix the defect

DMAIC: Analyze

- Analyze data collected
 - Identify gaps between current performance and goal performance
- Determine root causes of defects
 - Sources of variation
- Look for opportunities for improvements
 - Prioritize them

DMAIC: Analyze

- Example: Grocery Store
 - Horizontal bar graph showing percentages of defect occurrences



DMAIC: Analyze

- Major factors effecting process
 - Machines
 - Materials
 - Methods
 - Mother Nature
 - Measurements
 - People

DMAIC: Analyze

- Process analysis
 - Sub process Mapping
 - Start with High Level Process Map from Define phase
 - Reduce or eliminate inefficient steps
 - Analyze map for non-value added steps
 - Categorize non-value added steps
- Root cause analysis
 - Determine cause of defects
 - Open
 - Brainstorm all explanations of current sigma process
 - Narrow
 - Consolidate similar ideas and vote on most likely causes
 - Close
 - Test theories with data

DMAIC: Improve

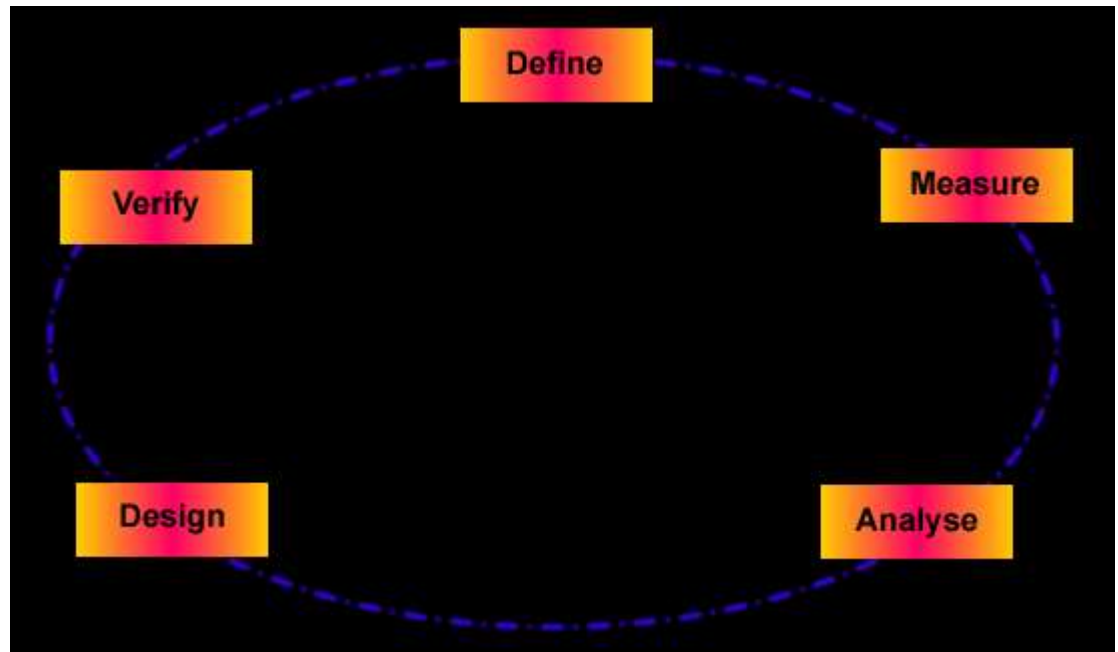
- Create innovative solutions to fix and prevent problems using technology and discipline
 - Create a solution for each verified root cause
- Select solutions
 - Implement solutions either individually or in groups
 - Recalculate sigma for each implementation

DMAIC: Control

- Control improvements to keep process on new course
 - Prevent reverting to “old way”
- Ensure any variances are corrected before they result in defects
 - Set up pilot runs
- Require the development, documentation and implementation of an ongoing monitoring plan
- Institutionalize the improvements through the modification of systems and structures (staffing, training, incentives, etc.)

DMADV

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First three steps are similar to those in DMAIC



DMADV-Design

- Design details
- Optimize design
- Run simulations if necessary
- Prepare for design verification

DMADV-Verify

- Verify design
- Set up pilot runs
- Implement process
- Train process owners
- Hand over to process owners

Other versions of DMADV

Methodology	Proponent
DCCDI (Define, Customer Concept, Design and Implement)	Geoff Tennant
CDOC (Conceptualize, Design, Optimize, Control)	SBTI
DCDOV* (Define, Concept, Design, Optimize, Verify) <i>*derived from SBTI CDOC roadmap</i>	Uniworld
DMADOV (Define, Measure, Analyze, Design, Optimize and Verify)	General Electric
DMEDI (Define, Measure, Explore, Develop and Implement)	PricewaterhouseCoopers
IDOV (Identify, Design, Optimize and Validate)	
I2DOV (Invent, Innovate, Develop, Optimize, Validate)	

Organizational Roles in Six Sigma

- Under a Six Sigma program, the members of an organization are assigned specific roles to play, each with a title. This highly structured format is necessary in order to implement Six Sigma throughout the organization. There are seven specific responsibilities or “role areas” in a Six Sigma program, which are as follows.
- ***Leadership***
- A leadership team or council defines the goals and objectives in the Six Sigma process. Just as a corporate leader sets a tone and course to achieve an objective, the Six Sigma council sets the goals to be met by the team. Here is the list of Leadership Council Responsibilities:
 - Defines the purpose of the Six Sigma program
 - Explains how the result is going to benefit the customer
 - Sets a schedule for work and interim deadlines
 - Develops a mean for review and oversight
 - Support team members and defend established positions

Organizational Roles in Six Sigma

- ***Sponsor***
- Six Sigma sponsors are high-level individuals who understand Six Sigma and are committed to its success. The individual in the sponsor role acts as a problem solver for the ongoing Six Sigma project. Six Sigma is generally led by a full-time, high level champion, such as an Executive Vice President. Sponsors are the owners of processes and systems, who help initiate and coordinate Six Sigma improvement activities in their areas of responsibilities.
- ***Implementation Leader***
- The person responsible for supervising the Six Sigma team effort, who supports the leadership council by ensuring that the work of the team is completed in the desired manner, is the Implementation Leader. Ensuring success of the implementation plan and solving problems as they arise, training as needed, and assisting sponsors in motivating the team are some of the key responsibilities of an implementation leader.
- ***Coach***
- Coach is a Six Sigma expert or consultant who sets a schedule, defines result of a project, and who mediates conflict, or deals with resistance to the program. Duties include working as a go-between for sponsor and leadership, scheduling the work of the team, identifying and defining the desired results of the project, mediating disagreements, conflicts, and resistance to the program and identifying success as it occurs.

Organizational Roles in Six Sigma

- ***Team Leader***
- It is an individual responsible for overseeing the work of the team and for acting as a go-between with the sponsor and the team members. Responsibilities include communication with the sponsor in defining project goals and rationale, picking and assisting team members and other resources, keeping the project on schedule, and keeping track of steps in the process as they are completed.
- ***Team Member***
- An employee who works on a Six Sigma project, given specific duties within a project, and has deadlines to meet in reaching specific project goals. Team members execute specific Six Sigma assignments and work with other members of the team within a defined project schedule, to reach specifically identified goals.
- ***Process Owner***
- The individual who takes on responsibility for a process after a Six Sigma team has completed its work.

Benefits of using Six Sigma

Organizations face rising costs and increasing competition every day. Six Sigma allows you to combat these problems and grow their businesses the following ways:

- ***Increases revenue***
 - Lean Six Sigma increases your organization's revenue by streamlining processes.
 - Streamlined processes result in products or services that are completed faster and more efficiently at no cost to quality.
 - Simply put, Lean Six Sigma increases revenue by enabling your organization to do more with less – Sell, manufacture and provide more products or services using fewer resources.

Benefits of using Six Sigma

- ***Decreases costs***

Six Sigma decreases your organization's costs by:

- Removing “Waste” from a process. Waste is any activity within a process that isn't required to manufacture a product or provide a service that is up to specification.
- Solving problems caused by a process. Problems are defects in a product or service that cost your organization money.

- Basically, Six Sigma enables you to fix processes that cost your organization valuable resources.

Benefits of using Six Sigma

- ***Improves efficiency***

Six Sigma improves the efficiency of your organization by:

- Maximizing your organization's efforts toward delivering a satisfactory product or service to your customers
- Allowing your organization to allocate resources/revenue produced from your newly improved processes towards growing your business

- Simply put, Six Sigma enables you to create efficient processes so that your organization can deliver more products or services, with more satisfied customers than ever before.

Benefits of using Six Sigma

- ***Develops effective people/employees***
Six Sigma develops effective employees within your organization by:
 - Involving employees in the improvement process. This promotes active participation and results in an engaged, accountable team.
 - Building trust. Transparency throughout all levels of the organization promotes a shared understanding of how each person is important to the organization's success.
- Basically, Six Sigma develops a sense of ownership and accountability for your employees. This increases their effectiveness at delivering results for any improvement project they are involved in. Quite often, this benefit is overlooked by organizations who implement Six Sigma, but it's underlying advantages dramatically increase the chances of continued success of Six Sigma, and your business.

Example of Six Sigma Calculation

A project is based on a billing mechanism. The team needs the customer to have the right bills sent. For this process, they have described one opportunity-either the bill is correct or not. In terms of complexity, all of the bills generated are the same.

The team took a sample of 250 bills and identified 60 defects.

The following example demonstrates how a Six Sigma calculation is performed using real process data.

Based on the Process Sigma table, the process being implemented by the team only has a sigma rating between 2 to 3. The higher the degree of sigma, the lower the number of defects.

Metric	Formula	Calculation	Result
DPU	Defects ÷ Total Units	$60 \div 250$	0.24
DPO	Defects ÷ (Total Units × Opportunities Per Unit)	$60 \div (250 \times 1)$	0.24
DPMO	$DPO \times 10^6$	0.24×10^6	240,000
Yield	$1 - DPO$	$1 - 0.24$	0.76

DMAIC: Measure

- Find the baseline σ
- Defects / Million opportunities =
Number of Defects x 1,000,000

Number of Units x Number of Opportunities

Process Sigma Short Term	Process Sigma Long Term	Yield	Defects per 1,000,000
6	4.5	99.99966	3
5.9	4.4	99.99946	5
5.8	4.3	99.99915	9
5.7	4.2	99.9987	13
5.6	4.1	99.9979	21
5.5	4	99.9968	32

Six Sigma Conversion Table

Yield	DPMO	Sigma	Yield	DPMO	Sigma	Yield	DPMO	Sigma
6.6%	934,000	0	69.2%	308,000	2	99.4%	6,210	4
8.0%	920,000	0.1	72.6%	274,000	2.1	99.5%	4,660	4.1
10.0%	900,000	0.2	75.8%	242,000	2.2	99.7%	3,460	4.2
12.0%	880,000	0.3	78.8%	212,000	2.3	99.75%	2,550	4.3
14.0%	860,000	0.4	81.6%	184,000	2.4	99.81%	1,860	4.4
16.0%	840,000	0.5	84.2%	158,000	2.5	99.87%	1,350	4.5
19.0%	810,000	0.6	86.5%	135,000	2.6	99.90%	960	4.6
22.0%	780,000	0.7	88.5%	115,000	2.7	99.93%	680	4.7
25.0%	750,000	0.8	90.3%	96,800	2.8	99.95%	480	4.8
28.0%	720,000	0.9	91.9%	80,800	2.9	99.97%	330	4.9
31.0%	690,000	1	93.3%	66,800	3	99.977%	230	5
35.0%	650,000	1.1	94.5%	54,800	3.1	99.985%	150	5.1
39.0%	610,000	1.2	95.5%	44,600	3.2	99.990%	100	5.2
43.0%	570,000	1.3	96.4%	35,900	3.3	99.993%	70	5.3
46.0%	540,000	1.4	97.1%	28,700	3.4	99.996%	40	5.4
50.0%	500,000	1.5	97.7%	22,700	3.5	99.997%	30	5.5
54.0%	460,000	1.6	98.2%	17,800	3.6	99.9980%	20	5.6
58.0%	420,000	1.7	98.6%	13,900	3.7	99.9990%	10	5.7
61.8%	382,000	1.8	98.9%	10,700	3.8	99.9992%	8	5.8
65.6%	344,000	1.9	99.2%	8,190	3.9	99.9995%	5	5.9
						99.99966%	3.4	6

Six Sigma Calculation Example

Step-by-Step Process Quality Calculation

Step 1 – Identify Defects

Defects Number of errors or failures found in the process.

Example: Defects = 25

Step 2 – Identify Units

Units Total number of items or services produced.

Example: Units = 500

Step 3 – Identify Opportunities per Unit

Opportunities Number of possible defect locations in each unit.

Example: Opportunities per Unit = 4

Step 4 – Calculate DPO

Defects per Opportunity

$$\text{DPO} = \frac{\text{Defects}}{\text{Units} \times \text{Opportunities per Unit}}$$

Example Calculation

$$\text{DPO} = 25 \div (500 \times 4)$$

$$\text{DPO} = 0.0125$$

Step 5 – Calculate Process Yield

$$\text{Yield} = 1 - \text{DPO}$$

$$\text{Example: Yield} = 1 - 0.0125 \rightarrow 0.9875 - 98.75\%$$



$$\text{DPO} = \text{Defects} \div (\text{Units} \times \text{Opportunities per Unit})$$

Six Sigma Process Improvement Flow



Measure Units, Defects, and Opportunities



Calculate DPU



Calculate DPO



Calculate DPMO



Determine Sigma Level



Identify Improvement Opportunities



Implement Process Improvements



Monitor and Sustain Gains